

SmartHire Gen AI Final Evaluation Report

1) Objective :

The objective of SmartHire GenAI is to help candidates manage their resumes and find suitable job opportunities using Generative AI.

The system allows users to:

- Create and save a candidate profile from their resume.
- Analyze the profile against job descriptions and suggest missing skills or improvements.
- Rewrite and improve the resume based on the selected job.
- Search and match jobs based on the candidate's skills and profile.
- Use an AI Career Mentor RAG Chat to get career guidance based on the available documents.

Overall, the goal is to provide personalized, AI-based support throughout the job-search process.

2) Retrieval relevance :

I tested the job-matching system using sample candidate profiles. The retrieved jobs were checked manually to determine whether they were relevant to the candidate's skills and profile. Relevant results were marked as “Yes” and irrelevant results as “No”. The hit rate was then calculated based on the number of relevant results.

Sample input (jobs and skills)	Relevant?
Data Analyst	Yes
Python Developer	Yes
Python	Yes
Digital Marketing	No
Product Manager	Yes
MATLAB	No
Graphic Designer	No
Java	Yes

Questions based on Semantic search :

- 1) Are the top retrieved jobs relevant to the candidate's profile? Yes
- 2) Does the search return relevant jobs within the top 3 results? Yes
- 3) How many of the top 5 jobs are relevant to the candidate? 2/3
- 4) Are the retrieved jobs related to the candidate's experience and target role? Yes

3) Answer Quality :

Question	Correct?	Grounded?	Helpful?
What skills are required for data analyst?	Yes	Yes	Yes
What are the key skills, market research methods, and tools required for product manager?	Yes	Yes	Yes
What are the core languages, system design skills, and technical focus areas required for a software engineer?	Yes	Yes	Yes
What is computer?	Yes	No	No
What are the core skills, spreadsheet tools, and visualization applications required for a data analyst?	Yes	Yes	Yes

4) Prompt comparison :

Basic Prompt :“Analyze the provided information and give a clear, relevant, and helpful answer. Use only the information provided and avoid making up facts.”

Improved Prompt :

```
CV_IMPROVEMENT_SYSTEM_PROMPT = ""
```

You are an expert career mentor and professional resume writer.

Your task is to analyze a candidate's resume against a target job description and provide specific, highly actionable feedback to help them land the job.

```
""
```

```
CV_IMPROVEMENT_USER_PROMPT = ""
```

Here is the candidate's parsed resume details:

```
{resume_json}
```

Here is the target job description:

```
{job_description}
```

Please provide your feedback in the following format:

1. ***Missing Skills***: List important skills mentioned in the job description that are missing from the resume.
2. ***Weak Bullet Points***: Identify 2-3 weak or vague bullet points in the experience section and explain how to improve them.
3. ***Rewritten Professional Summary***: Write an optimized, impactful professional summary tailored specifically for this target role.

```
""
```

```
Basic prompt :MENTOR_SYSTEM_PROMPT = ""
```

You are an expert AI assistant and career mentor.

Your task is to answer the user's questions clearly, accurately, and helpfully using the provided context.

If the answer is not available in the provided context, politely say that you do not have enough information to answer.

Do not make up information.

```
""
```

```
MENTOR_USER_PROMPT = ""
```

Use the following context to answer the user's question:

{context}

User Question:

{input}

Provide a clear, concise, and helpful answer. Use bullet points when appropriate.

"""

Improved Prompt :

MENTOR_SYSTEM_PROMPT = """

You are SmartHire, an expert AI career mentor.

Your task is to answer user queries strictly using the provided context regarding professional career roadmaps, skills, tools, and certifications (such as Cloud Analyst, Data Analyst, Full Stack Developer, Cybersecurity Analyst, and Cloud Architect).

If a question is outside this context or not related to these career paths, politely state that you cannot answer and specify your operational scope.

"""

MENTOR_USER_PROMPT = """

Context information from the SmartHire knowledge base:

{context}

Given the context information above, answer the following query accurately and concisely using bullet points where appropriate:

Query: {input}

"""

Conclusion :

The improved prompt produced a more focused and useful response compared to the basic prompt.

5) Hallucination check :

You need to test whether the AI Mentor makes up information when the answer is not available in your career documents.

Text Question	Expected Behavior	Result
Who is the current prime minister of India?	I am sorry, but I cannot answer that question as it is not covered in the provided context. I can only assist with questions regarding the provided career roadmaps (Cybersecurity Analyst, Data Analyst, and Full Stack Developer).	Yes  = Passed: The AI did not hallucinate and stayed within the provided career information.
What is a computer?	I am sorry, but I cannot answer that question as it is not covered in the provided context. I can only assist with questions regarding the provided career roadmaps (Cybersecurity Analyst, Data Analyst, and Full Stack Developer).	Yes  = Passed: The AI did not hallucinate and stayed within the provided career information.

Conclusion :

“The hallucination check verified whether the AI Mentor avoids generating information that is not available in the provided career notes.”

Overall Conclusion :

“The evaluation shows that SmarHire GenAI can provide relevant job results, useful career guidance, and improved responses using better prompts. The RAG-based mentor also helps reduce unsupported or incorrect answers by using the available career documents.”